



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

MDI 3003 – Advanced Predictive Analysis

Sarah Blessy N
23MID 0296

LAB Assessment 6

Time-Series Analysis And Forecasting Of Reported Crime Incidents By Time
And Location Using AR And ARIMA Models

From Incident-Level Public-Safety Records To Leakage-Safe Temporal Forecasting

List Of Contents

S.No	Topics	Page Num
1	Executive Summary	2
2	Business Problem	2
3	Dataset Description	3
4	Methodology	5
5	EDA and Feature Engineering	8
6	Model Development	10
7	Evaluation Results	11
8	Limitations And Risks	14
9	Future Scope	15
10	Conclusion	15

GitHub Repo → <https://github.com/blessy228/Course---Predictive-Analysis>

1. Executive Summary

This project focuses on forecasting the number of reported crime incidents in selected Chicago police districts. The main goal is to use past crime records to estimate future weekly crime levels. Such forecasts can help understand crime patterns and support better planning based on expected crime activity.

The analysis uses the Chicago Crimes - 2001 to Present dataset from the City of Chicago. The study uses crime records from 1 January 2019 to 31 December 2024 and focuses on two police districts: District 001 and District 012. The crime records are converted into weekly incident counts, with missing weeks filled with zero incidents.

The data is first checked and cleaned. Date values are converted into a standard date format, records with invalid dates are removed, and duplicate case numbers are checked and removed when present. The analysis also checks missing values and studies the weekly, monthly, and yearly crime patterns.

Three main forecasting approaches are compared: a Naive baseline, AutoRegressive (AR) model with 4 lags, and ARIMA. Several ARIMA models are tested, and the model with the lowest AIC is selected for each district. Model performance is measured using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The analysis also uses rolling validation to check model performance over different time periods.

The final forecasting model is selected separately for each district based on the ARIMA model comparison. The analysis also provides a 95% prediction interval and checks the model residuals using residual plots, residual autocorrelation, and the Ljung-Box test. An additional high/low crime classification is created using the training median as the threshold. The main limitation is that the forecasts represent reported crime incident counts, not the actual level of crime in the area. The code also focuses only on historical crime counts and does not include other factors that may affect crime levels. Therefore, the forecasts should be used as an estimate of reported crime activity rather than as a complete measure of actual crime risk.

2. Business Problem

2.1 Stakeholders

The main stakeholders are police departments and teams responsible for planning and managing police resources. The analysis focuses on two Chicago police districts, District 001 and District 012. The forecast can provide information about the expected number of reported crime incidents in these districts.

2.2 Business Decision

The main decision supported by this analysis is planning based on expected weekly crime levels. Historical crime data is used to forecast future reported crime incidents. These forecasts can help stakeholders understand whether the expected crime level is increasing or decreasing and support planning activities.

2.3 Target Variable

The target variable is the number of reported crime incidents per week.

The original crime records are grouped by week using the date field. Each week is represented by the total number of reported incidents. The analysis covers the period from 1 January 2019 to 31 December 2024.

2.4 Error Costs

The code evaluates forecasting errors using MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error).

- MAE shows the average size of the forecasting error.
- RMSE gives more importance to larger forecasting errors.

A forecasting error can result in an incorrect estimate of the expected crime level. However, the code does not assign a specific financial, operational, or social cost to false predictions. Therefore, exact error costs cannot be stated from the available analysis.

2.5 Intended Uses

The intended use of the model is to:

- Forecast weekly reported crime incidents.
- Understand historical crime patterns.
- Compare forecasting models.
- Support planning based on expected crime activity.
- Compare crime patterns between District 001 and District 012.

2.6 Prohibited Uses

- The code does not support using the forecast to determine actual crime prevalence or individual-level crime risk.
- The results should not be treated as a prediction of whether a particular person will commit a crime or whether a particular person is likely to become a victim. The model works with reported crime incident counts at the district level.
- The final code clearly states that the forecasts represent reported crime incident counts, not actual crime prevalence.

3. Dataset Description

3.1 Data Source

The project uses the Chicago Crimes - 2001 to Present dataset provided through the City of Chicago Socrata platform. The code accesses the dataset through its Socrata API endpoint.

3.2 Data Coverage

Although the dataset contains crime records from 2001 to the present, this project uses records from 1 January 2019 to 31 December 2024.

The analysis is limited to two police districts:

- District 001
- District 012

The original records are converted into weekly crime incident counts. The selected frequency is W-MON, meaning the data is organized on a weekly basis.

3.3 Dataset Schema

The code mainly uses the following fields:

Field	Use in the analysis
date	Date and time of the reported crime
district	Police district where the crime was reported
primary_type	Main type/category of crime
case_number	Used to identify duplicate records

The date field is converted into a date-time format. The district field is cleaned and formatted as a three-digit district code. The case_number is used to check for duplicate records.

3.4 Target Variable

The target variable is weekly reported crime incidents. The code counts the number of crime records for each week and creates a time series named incidents. Weeks without records are filled with zero. This weekly incident count is then used for forecasting

3.5 Data Quality

Several data quality checks are performed before modelling:

- Invalid or missing dates are removed.
- Duplicate case_number values are checked.
- Duplicate case records are removed when found.
- District values are cleaned and formatted consistently.
- Missing values are checked and visualized.
- The availability of the selected districts is verified before downloading the data.

The same cleaning process is applied to both District 001 and District 012



3.6 Dataset Summary

Item	Description
Dataset	Chicago Crimes – 2001 to Present
Source	City of Chicago Socrata dataset
Dataset ID	ijzp-q8t2
Analysis period	1 Jan 2019 – 31 Dec 2024

Locations	District 001 and District 012
Main date field	date
Location field	district
Crime category field	primary_type
Record identifier	case_number
Target	Weekly reported crime incidents
Time frequency	Weekly (W-MON)
Maximum records per district	200,000
Data cleaning	Invalid dates, duplicate cases, district formatting, missing-value checks
Main limitation	The target represents reported incidents, not actual crime prevalence

3.7 Dataset Limitations

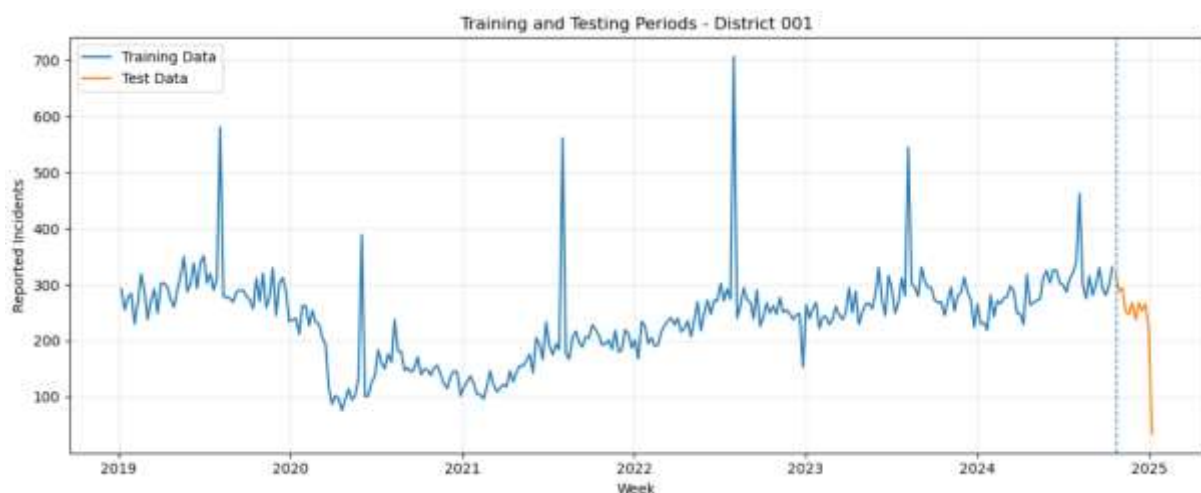
The analysis uses reported crime incidents, so the results do not represent the actual level of crime in an area. The code itself states that the forecasts represent reported crime incident counts and not actual crime prevalence.

The analysis also focuses on two districts and uses historical weekly incident counts for forecasting. Therefore, the results should be interpreted within the selected time period and locations.

4. Methodology

4.1 Data Split

The cleaned crime records are converted into weekly crime incident counts. The last 12 weeks are kept as the test set, while all earlier weeks are used for training. This split is performed separately for District 001 and District 012. This time-based split is used because the task is forecasting future crime incidents from past observations.



4.2 Preprocessing

The following preprocessing steps are performed:

1. Crime data is downloaded from the Chicago crime API.
2. The date field is converted into date-time format.
3. Records with invalid or missing dates are removed.
4. Duplicate case_number values are checked and removed when present.
5. District values are cleaned and converted into a standard three-digit format.
6. The data is grouped into weekly crime incident counts.
7. Missing weeks are added with a value of zero incidents.

The same preprocessing process is applied to both selected districts.

4.3 Forecasting Models

Three main forecasting approaches are compared.

Naive model: The last value from the training period is used as the prediction for all test weeks. This provides a simple baseline for comparison.

AR(4) model: An AutoRegressive model with 4 lags is used. This means the model uses previous observations to predict future crime counts.

ARIMA model: Four ARIMA configurations are tested:

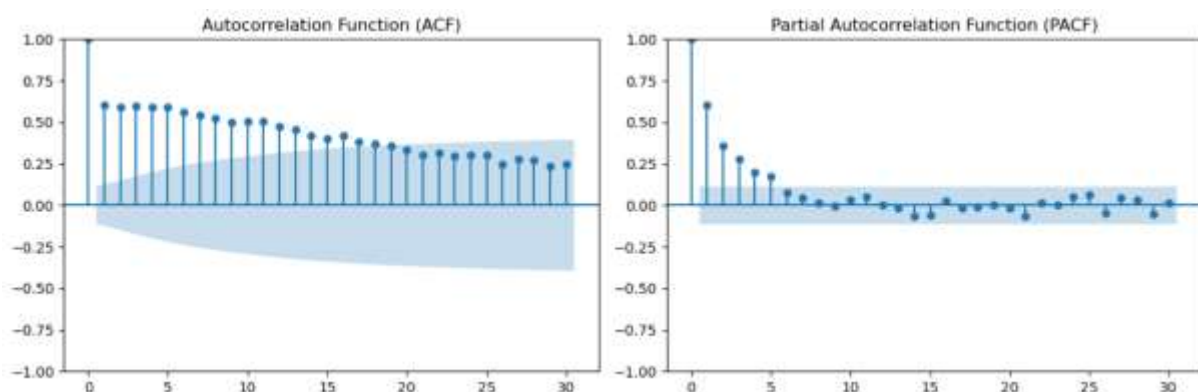
- ARIMA(1,0,0)
- ARIMA(2,0,0)
- ARIMA(1,1,1)
- ARIMA(2,1,1)

The model with the lowest AIC (Akaike Information Criterion) is selected as the best ARIMA model for each district. The selected ARIMA model is then used to forecast the 12 test weeks.

4.4 Stationarity and Time-Series Checks

Before model selection, the training data is tested using the Augmented Dickey-Fuller (ADF) test. The p-value is used to determine whether there is evidence of stationarity or non-stationarity. The analysis also uses:

- ACF (Autocorrelation Function) to study relationships between observations at different time lags.
- PACF (Partial Autocorrelation Function) to study the relationship at individual lags



4.5 Model Validation

Model performance is measured on the 12-week test period using:

- MAE: measures the average prediction error.
- RMSE: measures prediction error while giving more importance to larger errors.

The Naive, AR(4), and selected ARIMA models are compared using these two metrics.

The analysis also performs rolling-origin validation. Different earlier sections of the training data are used to train the ARIMA model, and the following 8 weeks are used for validation. The process moves forward by 8 weeks for each fold.

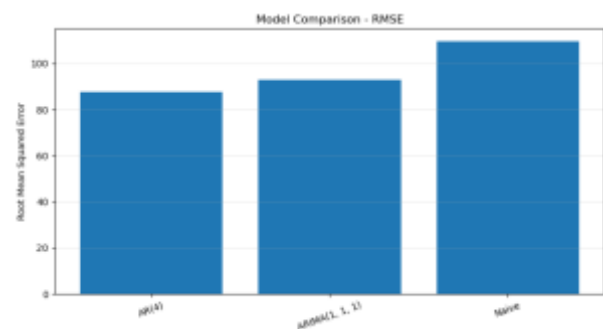
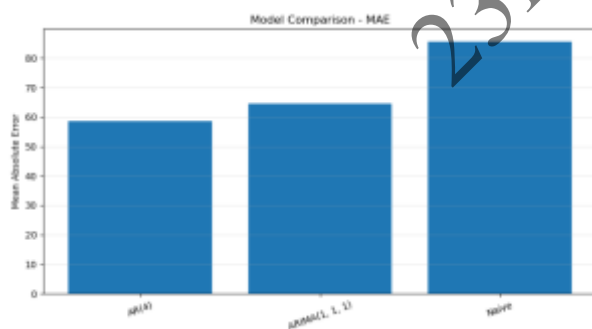
4.6 Model Tuning and Selection

ARIMA model tuning is performed by testing the four predefined ARIMA configurations. The model with the lowest AIC is selected. The selected model is then evaluated using the test data. The analysis also checks the model using:

- Prediction intervals
- Residual analysis
- Residual autocorrelation
- Ljung-Box test
- Rolling-origin validation

A separate SARIMA model is also tested for District 001 when enough data is available. The model uses:

- ARIMA order: (1,1,1)
- Seasonal order: (1,0,1,52)
- Seasonal period: 52 weeks



4.7 Reproducibility

The analysis uses a fixed random seed of 42. The output directory lab06_outputs is created automatically, and important results are saved as CSV and JSON files.

The saved outputs include:

- Model comparison results
- Test-set predictions
- Rolling-origin validation results
- Analysis manifest containing model and evaluation information

5. EDA and Feature Engineering

5.1. EDA and Feature Engineering

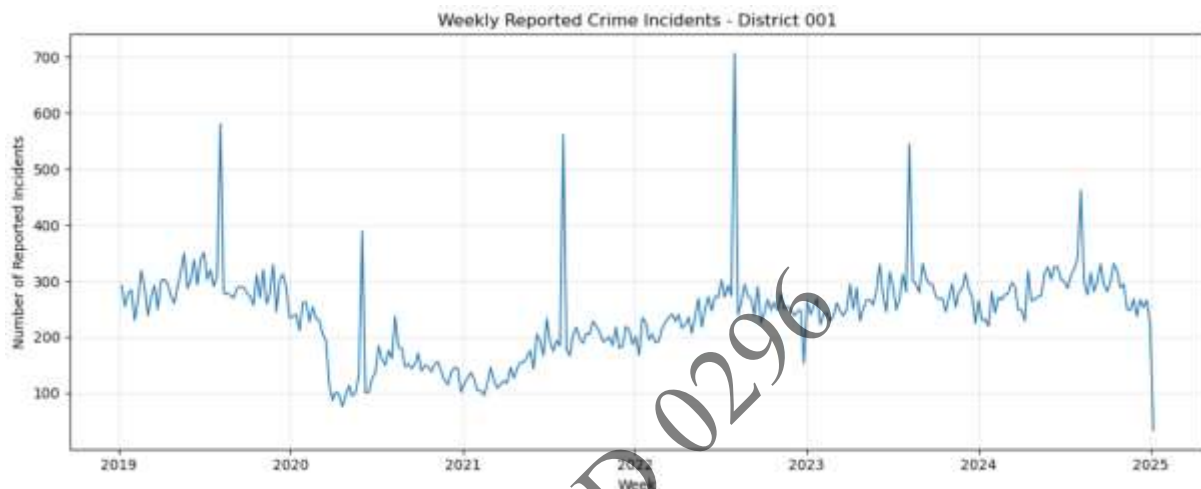
The exploratory data analysis (EDA) was focused on understanding the time pattern of reported crime incidents and deciding how the data should be prepared for forecasting.

1. Weekly crime counts were created

The individual crime records were converted into weekly incident counts. This created the main time-series target used by the forecasting models. Missing weeks were filled with zero incidents.

2. The overall weekly trend was checked

The weekly crime incident series was plotted to understand how reported crime changed over time. This was important because the project is a time-series forecasting problem.



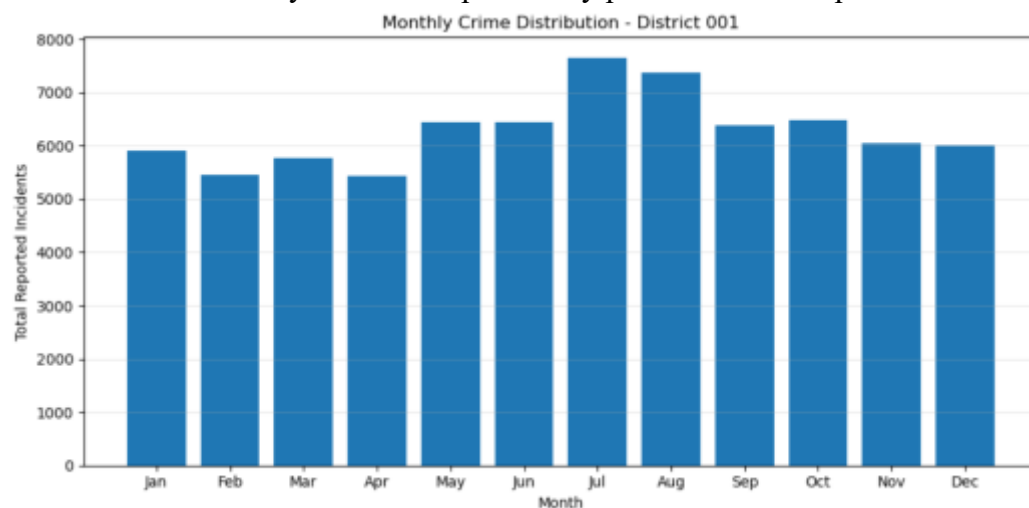
3. Short-term and longer-term trends were checked

A 4-week moving average and a 12-week moving average were calculated. These help show short-term and longer-term movement in reported crime counts.

Visualization: Rolling Crime Trend.

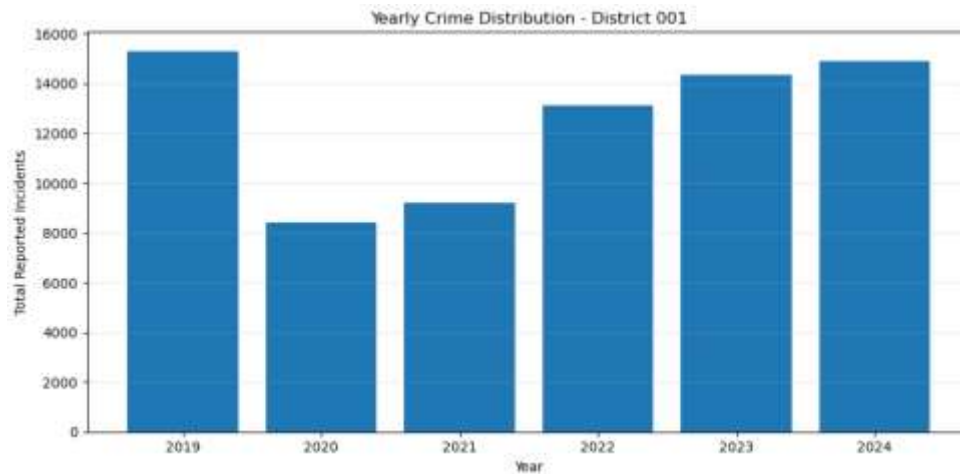
4. Monthly patterns were examined

Crime records were grouped by month to check whether reported crime levels differed across the months of the year. This helps identify possible seasonal patterns in the data.



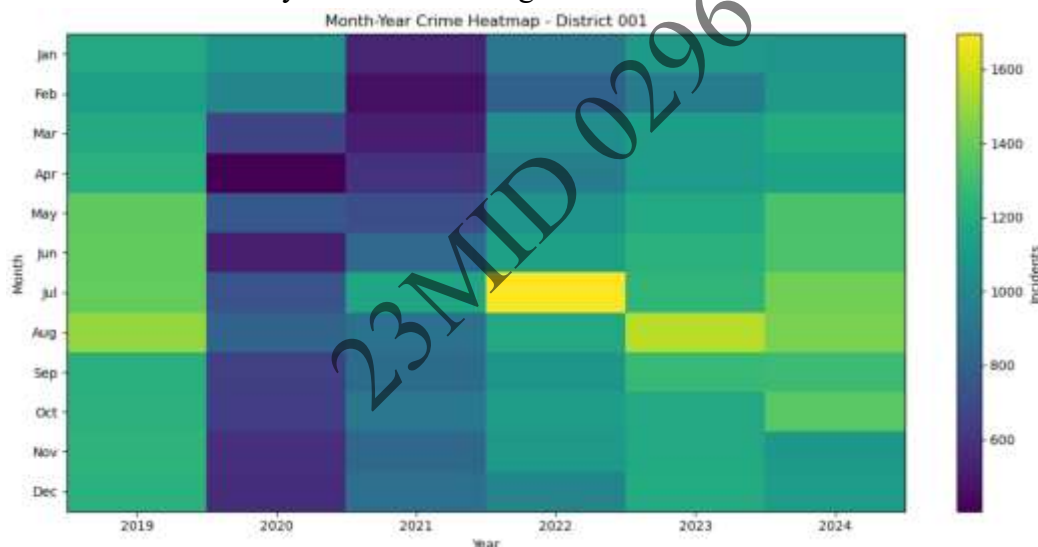
5. Yearly patterns were examined

Crime records were also grouped by year. This provides a simple view of how total reported incidents changed from year to year.



6. Month-year patterns were checked

A month-by-year table was created and displayed as a heatmap. This allows changes across both months and years to be seen together.



7. Stationarity was checked before forecasting

The Augmented Dickey-Fuller (ADF) test was applied to the training data. The p-value was used to determine whether there was evidence of stationarity or non-stationarity. This information is important for selecting an appropriate ARIMA structure.

8. Lag relationships were studied

ACF and PACF plots were created to study relationships between current crime counts and previous observations. These plots support the choice of time-series model and lag structure. Visualization: ACF and PACF plots.

9. EDA decisions were used in model development

The EDA supports treating the problem as a time-series forecasting task. The data is kept in time order, the final 12 weeks are reserved for testing, and AR, ARIMA, and SARIMA

models are evaluated. The seasonal period of 52 weeks is used for the SARIMA model to represent an annual weekly cycle when enough data is available.

5.2. Key EDA Findings Used for Modeling

EDA Area	Purpose	Modeling Decision
Weekly crime trend	Understand changes over time	Use time-series models
4-week and 12-week averages	Understand short- and long-term trends	Helps interpret changing crime levels
Monthly and yearly patterns	Check possible repeated patterns	Supports seasonal model testing
ACF	Check relationships with previous values	Supports AR/ARIMA models
PACF	Check individual lag relationships	Supports lag-based modelling
ADF test	Check stationarity	Supports ARIMA model selection
Weekly aggregation	Convert records into a consistent time series	Use weekly incident count as target

6. Model Development

1. Baseline model

A Naive forecasting model is used as the baseline. It uses the last observed training value as the forecast for all future test weeks. This gives a simple reference point for comparing the other models.

2. Comparative AR model

An AutoRegressive (AR) model with 4 lags is developed. It uses the previous four weekly crime counts to make predictions. The value of `ar_lags` is set to 4 in the configuration.

3. ARIMA models

Four ARIMA configurations are tested:

- ARIMA(1,0,0)
- ARIMA(2,0,0)
- ARIMA(1,1,1)
- ARIMA(2,1,1)

These models are compared to identify a suitable model for the weekly crime data.

4. Hyperparameter selection

The ARIMA models are selected using AIC. For each district, the configuration with the lowest AIC is selected as the best ARIMA model. This selection is performed separately for District 001 and District 012.

5. Seasonal model

A SARIMA(1,1,1)(1,0,1,52) model is also tested for District 001 when enough weekly data is available. The value 52 represents the seasonal period used in the code.

6. Experiment design

The data is kept in time order. The final 12 weeks are used as the test period, while the earlier observations are used for training. This same setup is used when comparing the Naive, AR(4), and selected ARIMA models.

7. Fair model comparison

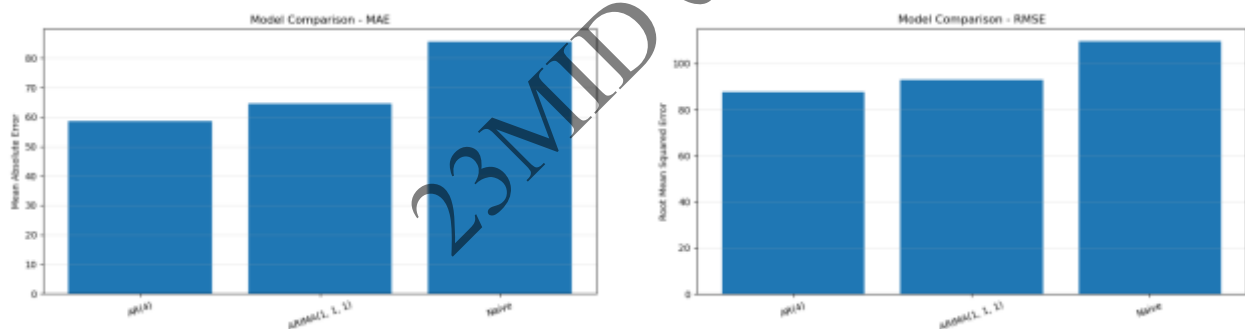
All main models are evaluated on the same test period using the same metrics: MAE and RMSE. This makes the comparison consistent because each model is tested against the same actual weekly crime counts.

8. Additional validation

Rolling-origin validation is used as an additional experiment. The model is trained using earlier observations and tested on the next 8 weeks, with the process repeated by moving forward 8 weeks at a time. This checks whether the selected model performs consistently across different time periods.

9. Final model evaluation

After model selection, the selected ARIMA model is used to forecast the test period. Prediction intervals and residual checks are also performed. This includes residual plots, residual ACF, and the Ljung-Box test to check whether important patterns remain in the model errors.



7. Evaluation Results and Discussion

7.1 Test Set Results

The final 12 weeks were used as the test period. The Naive, AR(4), and ARIMA models were evaluated using MAE and RMSE.

District	Model	MAE	RMSE
District 001	Naive	85.58	109.60
District 001	AR(4)	58.61	87.63
District 001	ARIMA(1,1,1)	64.50	92.90
District 012	Naive	54.50	87.07
District 012	AR(4)	58.97	92.77
District 012	ARIMA(1,1,1)	59.34	92.60

For District 001, the AR(4) model gives the lowest MAE and RMSE. It performs better than both the Naive and ARIMA models. For District 012, the Naive model gives the lowest MAE and RMSE compared with the AR(4) model. The ARIMA model has a slightly lower RMSE than AR(4), but its MAE is higher than the Naive model.

7.2 Rolling-Origin Validation

Rolling-origin validation was performed using 18 validation folds. Each fold uses an 8-week validation period.

District	Mean MAE	Median MAE	Mean RMSE	Median RMSE
District 001	29.75	24.16	42.36	32.15
District 012	28.80	24.76	34.28	30.85

The rolling validation results show that the model errors change across different time periods. For District 001, the RMSE reaches a high value in some folds, mainly because of larger errors in those periods. District 012 shows more stable RMSE values across the validation folds.

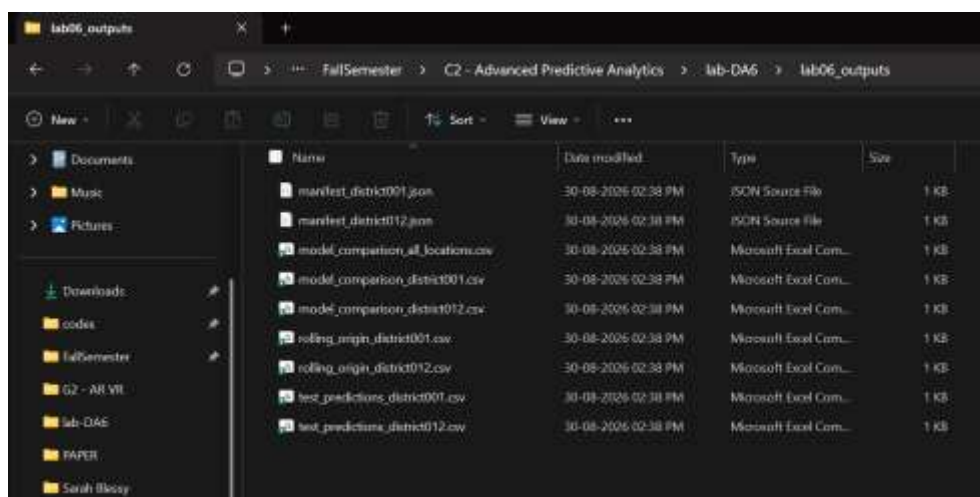
This shows that model performance can change depending on the time period being predicted. Therefore, using only one test period may not fully represent model performance.

7.3 Actual vs Predicted Results

For District 001, the actual weekly crime count changes from 318 incidents at the start of the test period to 34 incidents in the final week. The AR(4) predictions remain around 299–306 incidents, while the ARIMA predictions remain around 308 incidents. The Naive model gives a constant prediction of 331 incidents.

For District 012, the actual crime count starts at 351 incidents and falls to 47 incidents in the final week. The Naive model gives a constant prediction of 316 incidents. The AR(4) and ARIMA models remain around 322–331 and 326–327 incidents respectively.

The large fall in the final weeks shows that all three models have difficulty responding quickly to the sudden decrease in reported incidents.



7.4 Error Behaviour

The District 001 results show that AR(4) reduces the MAE from 85.58 for the Naive model to 58.61. RMSE also decreases from 109.60 to 87.63. This shows a clear improvement over the baseline.

For District 012, the Naive model has the lowest MAE at 54.50 and the lowest RMSE at 87.07. The AR(4) and ARIMA models do not improve the baseline on MAE.

The results show that a more complex model does not always give better forecasting performance. The simple Naive model performs best for District 012.

7.5 Model Performance Discussion

District 001: AR(4) is the best-performing model based on both MAE and RMSE. It uses previous weekly observations and provides better predictions than the baseline and ARIMA model.

District 012: The Naive model performs best based on both MAE and RMSE. This means that using the most recent observed value provides a strong baseline for this district during the test period.

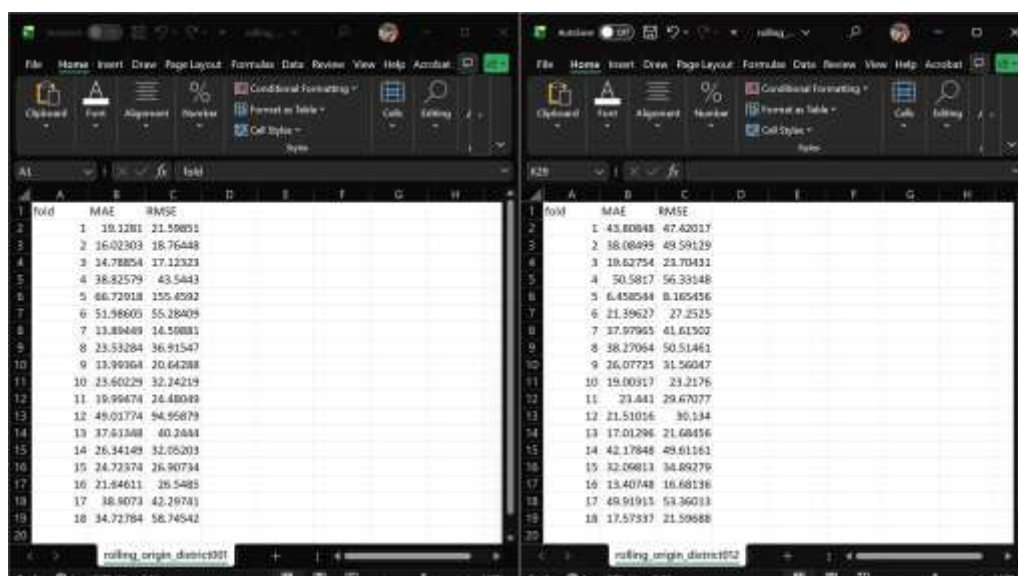
Therefore, there is no single model that performs best for both districts.

7.6 Rolling Validation Discussion

The rolling validation results support the need for testing model performance across multiple time periods. District 001 has a mean RMSE of 42.36, while District 012 has a lower mean RMSE of 34.28. However, individual folds show that errors can increase considerably during some periods.

For District 001, the fifth fold has an RMSE of 155.46, which is much higher than most other folds. For District 012, the highest RMSE is 56.33 in the fourth fold.

This indicates that forecasting performance is not constant over time and that sudden changes in crime counts can create larger prediction errors.



fold	MAE	RMSE
1	10.1281	21.50851
2	16.02303	18.76448
3	34.78854	17.12323
4	36.82579	43.5443
5	46.72018	155.4592
6	51.96603	55.18409
7	13.89449	14.50881
8	23.53284	36.91347
9	13.99364	20.64288
10	25.60229	32.24219
11	19.99474	24.48049
12	49.03774	94.95879
13	37.61348	40.2484
14	26.34149	32.05203
15	24.72374	26.90734
16	21.84611	26.5485
17	38.9073	42.29743
18	34.72784	58.74542

fold	MAE	RMSE
1	43.80848	47.42017
2	38.08495	49.59129
3	19.62754	23.70431
4	50.5817	56.33148
5	6.458544	8.165456
6	21.39627	27.2523
7	37.97903	41.61302
8	38.17064	50.51461
9	26.07725	31.56047
10	19.00317	23.2176
11	23.441	29.67077
12	21.51016	30.134
13	17.01296	21.68456
14	42.17848	49.61161
15	32.06813	34.89276
16	13.40748	16.68138
17	49.91913	53.36013
18	17.57337	21.59888

7.7 Overall Discussion

The evaluation shows that historical weekly crime counts can be used to create short-term forecasts, but model performance depends on the district and time period.

The main findings are:

- AR(4) is the best test-set model for District 001.
- Naive is the best test-set model for District 012.
- ARIMA(1,1,1) does not give the lowest MAE for either district.
- Rolling validation shows that model errors change across time.
- Large sudden changes in actual crime counts are difficult for the models to capture.
- A more complex forecasting model does not always perform better than a simple baseline

8. Limitations and Risk

1. Limited coverage

The analysis covers only District 001 and District 012 and uses data from 1 January 2019 to 31 December 2024. Therefore, the results cannot automatically be applied to other Chicago districts or to periods outside the selected data range.

2. Reported-crime bias

The target is the number of reported crime incidents, not the actual number of crimes that occurred. Differences in reporting can affect the recorded counts. Therefore, the forecasts should not be treated as a direct measure of actual crime prevalence.

3. Possible changes over time (drift)

Crime patterns can change over time. The rolling validation results show that model errors are not constant across all periods. For example, District 001 has a much higher RMSE in one validation fold (155.46) compared with most other folds. This suggests that some time periods are harder to forecast than others.

4. Unusual changes are difficult to predict

The test data contains a large fall in reported incidents near the end of the test period. For District 001, the actual weekly count falls from 318 to 34, while the model predictions remain much higher. District 012 also falls from 351 to 47. This shows that the models may not react quickly to sudden changes in crime activity.

5. Forecast uncertainty

The model provides 95% prediction intervals, but forecasts are still estimates and contain uncertainty. The interval gives a range of possible future values rather than a guaranteed result. Therefore, forecasts should be treated as decision-support information and not as exact future crime counts.



6. Invalid uses

The model should not be used to predict whether an individual person will commit a crime, identify individual offenders, or determine individual-level crime risk. The model forecasts weekly reported crime counts at district level. It should also not be used as a replacement for human decision-making or as proof of actual crime prevalence.

9. Future Improvements

1. Add better data and improve validation

The first priority is to include more useful data and test the models using stronger time-based validation. Future versions can include additional crime-related information and wider district coverage. Spatial and temporal splits can also be used to check whether the model works across different locations and time periods. Prediction intervals and other uncertainty measures should be monitored along with MAE and RMSE.

2. Build deployment and monitoring

The forecasting system can be developed into a regular monitoring system that updates forecasts as new crime records become available. Model performance, prediction errors, and changes in crime patterns should be monitored over time. If forecast errors increase or the data pattern changes, the model should be reviewed and retrained. This will make the system more suitable for continued operational use.

10. Conclusion

The project successfully developed a time-series forecasting approach for weekly reported crime incidents in Chicago Police Districts 001 and 012. The analysis followed a consistent process of data cleaning, exploratory analysis, model development, testing, and rolling validation.

The stated success criteria were to compare forecasting models using MAE and RMSE, validate their performance over time, and select a suitable model for each district.

Based on the evaluation already presented:

- District 001: The AR(4) model achieved the best test-set performance, with the lowest MAE and RMSE among the compared models.
- District 012: The Naive model achieved the best test-set performance, showing that a simple baseline can perform better than more complex models for this district.
- Rolling-origin validation showed that model errors change across different time periods.
- The models had difficulty capturing sudden changes in reported crime counts.

Therefore, the model meets the basic success criteria for comparing and forecasting weekly reported crime counts, but it should be treated as a decision-support tool rather than an exact prediction system. The results are limited to the selected districts and historical period, and the target represents reported crime rather than actual crime prevalence